force. But let’s now think about this interpretation a little more closely. Mathematically, the curvature of spacetime—like the curvature of the trampoline—reflects the distorted distance relations between its points. Physically, the gravitational force felt by an object is a direct reflection of this distortion. In fact, by making the object smaller and smaller, the physics and the mathematics align ever more precisely as we get closer and closer to physically realizing the abstract mathematical concept of a point. But string theory limits how precisely Riemann’s geometrical formalism can be realized by the physics of gravity, because there is a limit to how small we can make any object. Once you get down to strings, you can’t go any further. The traditional notion of a point particle does not exist in string theory—an essential element in its ability to give us a quantum theory of gravity. This concretely shows us that Riemann’s geometrical framework, which relies fundamentally upon distances between points, is modified on microscopic scales by string theory.

This observation has a very small effect on ordinary macroscopic applications of general relativity. In cosmological studies, for example, physicists routinely model whole galaxies as if they are points, since their size, in relation to the whole of the universe, is extremely tiny. For this reason, implementing Riemann’s geometrical framework in this crude manner proves to be a very accurate approximation, as evidenced by the success of general relativity in a cosmological context. But in the ultramicroscopic realm, the extended nature of the string ensures that Riemann’s geometry simply will not be the right mathematical formalism. Instead, as we will, now see, it must be replaced by the quantum geometry of string theory, leading to dramatically new and unexpected properties.

A Cosmological Playground

According to the big bang model of cosmology, the whole of the universe violently emerged from a singular cosmic explosion, some 15 or so billion years ago. Today, as originally discovered by Hubble, we can see that the “debris” from this explosion, in the form of many billions of galaxies, is still streaming outward. The universe is expanding. We do not know whether this cosmic growth will continue forever or if there will come a time when the expansion slows to a halt and then reverses itself, leading to a cosmic implosion. Astronomers